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Bachelor of Computer Applications

Semester – V

Deep Learning

Title: Comparison of Deep Learning Architectures

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Date:31/08/2026

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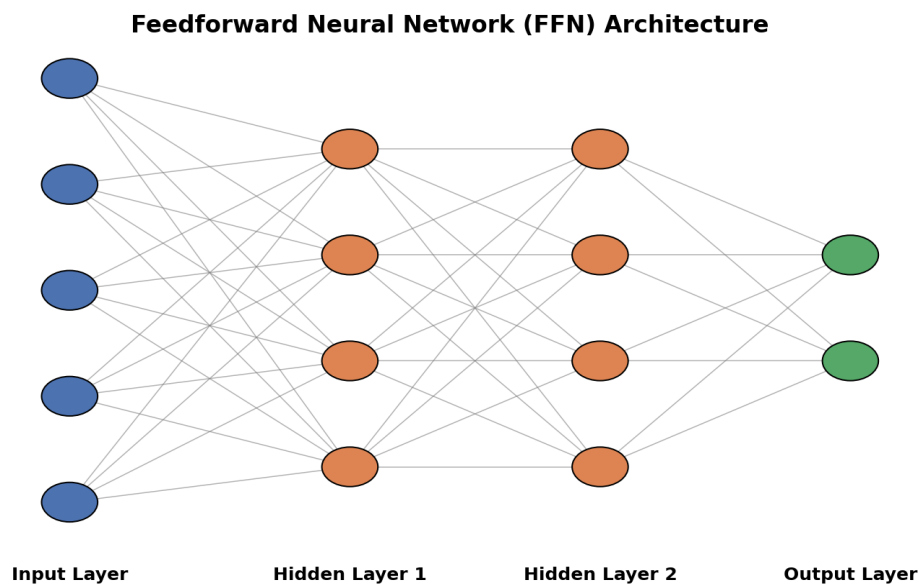
Introduction

Deep Learning has become the driving force behind modern Artificial Intelligence, powering applications that range from image recognition to language translation. Different tasks call for different network designs, and understanding these designs is essential to choosing the right tool for a given problem. This report studies and compares three foundational deep learning architectures: the Feedforward Neural Network (FFN), the Convolutional Neural Network (CNN), and the Recurrent Neural Network (RNN). For each architecture, the basic structure is explained, suitable real-world applications are identified, and the three are compared side by side based on structure, input data, and applications.

Task 1: Basic Architecture of Each Network

1.1 Feedforward Neural Network (FFN)

A Feedforward Neural Network is the simplest type of artificial neural network. It consists of an input layer, one or more hidden layers, and an output layer. Information moves strictly in one direction — from the input layer, through the hidden layers, to the output layer — with no cycles or loops. Each neuron in a layer is typically connected to every neuron in the next layer (a “fully connected” or “dense” layer), and each connection carries a weight that is adjusted during training through backpropagation.



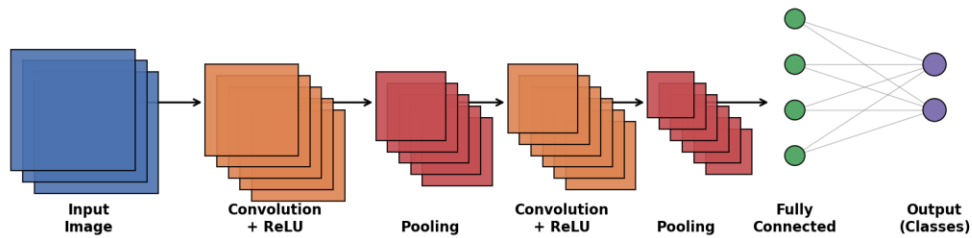
Key characteristics:

- No memory of past inputs — every prediction is independent of previous ones.
- Works well with fixed-size, structured (tabular) input data.
- Simple to design and computationally efficient to train.

1.2 Convolutional Neural Network (CNN)

A Convolutional Neural Network is designed to process data that has a grid-like topology, such as images. Instead of connecting every neuron to every input value, a CNN uses small filters (kernels) that slide across the input to detect local patterns such as edges, textures, and shapes. A typical CNN is composed of alternating convolutional layers and pooling layers, followed by one or more fully connected layers that perform the final classification or regression.

Convolutional Neural Network (CNN) Architecture



Key characteristics:

- Convolutional layers extract spatial features while preserving spatial relationships.
- Pooling layers reduce dimensionality and make the network more robust to small shifts in the input.
- Parameter sharing (the same filter is reused across the image) makes CNNs efficient for large inputs like images.

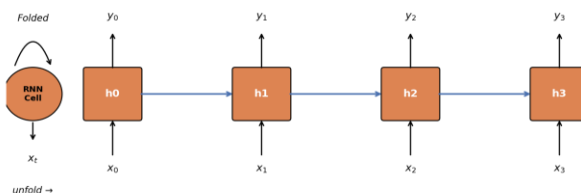
Applications:

- Image classification
- Face recognition
- Object detection
- Medical image analysis
- Handwritten digit recognition

1.3 Recurrent Neural Network (RNN)

A Recurrent Neural Network is designed for sequential data, where the order of inputs matters. Unlike FFNs and CNNs, an RNN has connections that loop back on themselves, allowing information to persist from one step in the sequence to the next through a “hidden state.” At each time step, the network combines the current input with the hidden state carried over from the previous step, giving the network a form of memory.

Recurrent Neural Network (RNN) Architecture — Unfolded Through Time



Key characteristics:

- Maintains a hidden state that acts as memory of previous inputs in the sequence.
- Well suited to variable-length sequential data such as text or time series.
- Can suffer from vanishing or exploding gradients over long sequences; advanced variants such as LSTM and GRU address this limitation.

Applications:

- Speech recognition
- Text generation
- Language translation
- Sentiment analysis
- Time-series forecasting

Task 2: Suitable Applications for Each Architecture

2.1 Feedforward Neural Network Applications

- Predicting numerical outcomes from structured/tabular data (e.g., house price prediction).
- Basic classification tasks such as credit risk scoring or customer churn prediction.
- Simple pattern recognition problems where input size is fixed and there is no spatial or temporal structure.

2.2 Convolutional Neural Network Applications

- Image classification (e.g., identifying objects, animals, or handwritten digits).
- Object detection and facial recognition in photos and video.
- Medical image analysis, such as detecting tumors in X-rays or MRI scans.
- Self-driving car perception systems (lane detection, obstacle recognition).

2.3 Recurrent Neural Network Applications

- Natural language processing tasks such as machine translation and text generation.
- Speech recognition and voice assistants.
- Time-series forecasting, such as stock price or weather prediction.
- Music generation and other sequence-modeling tasks.

Architecture	Suitable Input Data	Common Applications
FNN	Structured / tabular data	Classification, regression, sales prediction, fraud detection
CNN	Images / spatial data	Image recognition, object detection, medical imaging
RNN	Sequential / time-series data	Speech recognition, text analysis, translation, forecasting

Task 3: Comparison of Architectures

The table below summarizes the three architectures based on their structure, how data flows through the network, the type of input data they are best suited for, and their typical real-world applications.

Aspect	Feedforward Neural Network (FFN)	Convolutional Neural Network (CNN)	Recurrent Neural Network (RNN)
Structure	Fully connected layers; signals move in one direction from input to output with no cycles	Convolutional and pooling layers that extract spatial features, followed by fully connected layers	Neurons with feedback loops; hidden state is passed from one time step to the next
Data Flow	One-directional (forward only), no memory of previous inputs	One-directional, but preserves spatial relationships through local receptive fields	Sequential and cyclical; output depends on both current input and previous hidden state
Input Data	Fixed-size numerical or tabular vectors	Grid-like data such as images and video frames	Sequential data such as text, speech, or time series
Memory	None	None (limited spatial context only)	Yes — maintains a hidden state across time steps
Parameter Sharing	No	Yes (filters/kernels shared across the input)	Yes (same weights reused at every time step)
Typical Applications	Classification/regression on tabular data, basic pattern recognition	Image classification, object detection, facial recognition	Language modeling, speech recognition, time-series forecasting
Key Strength	Simplicity and fast training for structured data	Excellent at capturing spatial hierarchies and local patterns	Excellent at capturing temporal/sequential dependencies
Key Limitation	Cannot capture spatial or sequential structure	Not naturally suited for sequential/temporal data	Prone to vanishing gradients over long sequences; slower to train

Real-World Examples

FFN – Student Performance Prediction

Student information such as attendance, study hours, and previous marks can be used as inputs to predict student performance.

CNN – Face Recognition

A CNN can process an image and learn facial features to identify or verify a person.

RNN – Sentiment Analysis

An RNN can process words in a sentence sequentially and use previous words to determine whether the sentiment is positive, negative, or neutral.

Conclusion

Each deep learning architecture is optimized for a different type of problem. Feedforward Neural Networks offer a simple, general-purpose starting point for structured data. Convolutional Neural Networks excel at extracting spatial features from images and visual data, making them the standard choice for computer vision. Recurrent Neural Networks are built to handle sequential and time-dependent data, making them well suited

for language, speech, and time-series tasks. Understanding these differences allows practitioners to select the architecture that best matches the structure of their data and the nature of their problem.

Expected Output / Result

The three deep learning architectures—FNN, CNN, and RNN—were studied and compared based on their architecture, input data, working principles, and real-world applications. The project demonstrates how each architecture is suited to structured, spatial, or sequential data.

References

- Deep Learning course notes and classroom materials.
- Goodfellow, Bengio & Courville, Deep Learning.
- Neural network and deep learning educational resources.